



Who moves first? An intensity-based measure for information flows across stock exchanges

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ABSTRACT

In this paper we propose an innovative measure for information flows between stock exchanges. We develop an intensity-based information share using Russell's (1999) autoregressive conditional intensity model. Thereby we maintain the irregular nature of financial high frequency data and use durations and timing of price changes to determine the informationally dominant market. From our empirical application to US-listed Canadian stocks we conclude that the home market mostly reflects information first. On the basis of a cross-sectional analysis we find a positive correlation between the intensity-based information share and liquidity.

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1. Introduction

The last decade has seen increasing globalization and improved technology concerning the trading of financial assets. This development has sharpened competition among stock exchanges around the world, in particular where exchanges compete for the same crowd of investors, as in the case of cross-listed stocks. In order to attract investors, exchanges have to provide an efficient trading environment, where information is incorporated into prices as quickly as possible. This renders the contributions of each trading venue to the price discovery process of the common stock a quality feature.

This paper contributes to the current literature by proposing an alternative price discovery measure (*information share*). Our information share is based on Russell's (1999) autoregressive conditional intensity model (ACI). This bivariate intensity approach accounts for the informational content of time between consecutive quoted price changes within a market as well as the timing interdependencies between the price processes on both markets. In contrast to the commonly applied Hasbrouck (1995) methodology, we exploit the irregular occurrence of price changes and deliver a unique information share. We apply our new measure to

Canadian stocks that are traded on the Toronto Stock Exchange (TSX) and cross-listed on the New York Stock Exchange (NYSE). Furthermore, we examine potential determinants of information shares in a cross-sectional analysis.

Evidence from previous studies suggests that price discovery for cross-listed stocks mainly takes place in the home market. Eun and Sabherwal (2003) examine a sample of US-listed Canadian stocks based on the relative adjustment of prices in a market to deviations from the equilibrium price. They conclude that the contribution of the US market cannot be neglected, but that the home market clearly dominates price discovery. Adjustment coefficients as a measure of price discovery, however, have been criticized, since they do not account for the contemporaneous correlations and variances of markets' price innovation (see De Jong, 2002; Baillie et al., 2002). The majority of empirical studies apply Hasbrouck's (1995) method, in which the information share is defined as the contribution of a market's price innovation to the variance of the efficient underlying price innovations. Grammig et al. (2005), Hupperets and Menkveld (2002) and Korczak and Phylaktis (2010) use the Hasbrouck (1995) methodology to estimate the home and foreign market share in price discovery for US-listed stocks from various countries. They conclude that trading on the home market stock exchanges contributes most to price discovery, while trading on the NYSE primarily takes place to offset arbitrage opportunities.

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The main drawback of the Hasbrouck (1995) approach is that it merely delivers upper and lower bounds for an information share. The method requires equidistant sampled quotes and, depending on the chosen sampling frequency, the information share bounds can diverge considerably. Consequently, the conclusions concerning the leading market are rather vague (see Hupperets and Menkveld (2002) and Korczak and Phylaktis (2010)).¹ We take an alternative view on how to quantify the informational linkage between different exchanges trading the same underlying asset which is motivated by the following considerations. First, as acknowledged in the financial markets literature (e.g. Dufour and Engle, 2000; Engle, 2000; Engle and Lunde, 2003; Frijns and Schotman, 2009), the irregular occurrence of trades and quotes and the time between consecutive financial market events reveal the dynamics of price responses to new information in the market. Hence, arbitrary sampling schemes used to obtain regularly spaced data neglect this aspect of the price dynamics and induce an undesirable loss of information. Second, as pointed out by Hasbrouck (1995), "... the information share measures *who moves first* in the process of price adjustment." If price discovery is indeed understood as which market "moves first", an information share measure that is derived from a model that directly accounts for the irregular sequence and timing of the price process seems to be a straightforward approach. We exploit the irregularity of high frequency stock market data and propose an alternative information share by modeling the arrival rates (intensities) of the price processes using Russell's (1999) ACI model. Within this framework the dynamics of the intensity functions are driven by innovations that allow for a flexible interaction and simultaneously affect the conditional intensities in both markets.

In the case of cross-listed securities the law of one price applies and deviations from the equilibrium price can only be transitory. Consequently, when prices in both markets deviate, adjustment has to take place to restore the equilibrium. The main question is which market moves first and which markets adjust to restore the equilibrium price? If the market event that caused the deviation was due to new information, the arbitrage relation between prices in parallel markets forces an immediate incorporation of new information into the other market's price thereby increasing its intensity. This translates into positive spillover effects due to the innovations. We argue that the larger the effect of an innovation in one market on the other market's intensity, the greater is its intensity-based information share, i.e. the information flow that originates from this trading venue.

We empirically analyze the price discovery process of Canadian stocks traded on the TSX and cross-listed on the NYSE. Our results show the TSX to be a clear leader in the price discovery process. With an average information share of 73%, the contribution of the TSX is more pronounced than indicated by Eun and Sabherwal (2003).

We also examine potential determinants of a market's contribution to price discovery by conducting a cross-sectional regression of the intensity-based information share on stock-specific factors and liquidity-related variables.

Eun and Sabherwal (2003), Korczak and Phylaktis (2010), Grammig et al. (2005) and Frijns et al. (2010) among others analyze the determinants of price discovery in the home and foreign market and find a positive relation between the contributions to price discovery and liquidity. In line with this research, our results show that liquidity proxies are the only variables that are related to a market's price determination. This implies that providing an effi-

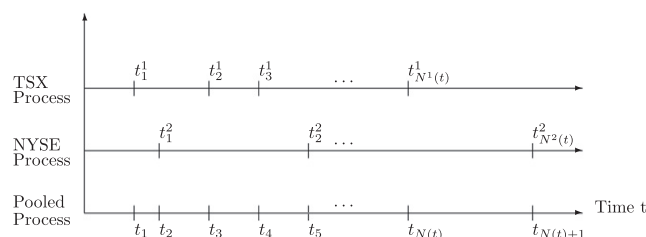


Fig. 1. Pooled point process illustration. The figure gives an illustration of a simple point process $N(t)$ that consists of two individual counting processes, $N^1(t)$ and $N^2(t)$. $\{t_i^1\}_{i=1}^n$ denotes the arrival times of events on the TSX and $\{t_i^2\}_{i=1}^n$ corresponds to price event times on the NYSE. A time sequence $\{t_i\}_{i=1}^n$ containing both event time series is obtained by pooling and ordering the individual event times. As a consequence, an event occurring on the TSX not only depends on its own history, but depends on the history of the NYSE process as well, and vice versa.

cient and liquid trading environment is of particular interest for small national stock exchanges that seek to maintain their dominance in the price discovery process of cross-listed stocks.

The remainder of the paper is organized as follows. Section 2 outlines the methodological details of the ACI model, describes the data and introduces the intensity-based information share. In Section 3 we present estimation results and discuss the estimated information shares. A cross-sectional regression analysis is provided in Section 4. Section 5 concludes.

2. A new measure for contributions to price discovery

2.1. The autoregressive conditional intensity model

In our empirical application we consider stocks that are simultaneously traded on the TSX and the NYSE. This section introduces a bivariate autoregressive conditional intensity (ACI) model that is applied to the price events² of a stock listed on both markets. We start by defining the point process $\{t_i^s\}_{i=1}^n$ as the stochastic sequence of price changes on the market s in calendar time t , where $s = 1$ corresponds to a price event on the TSX and $s = 2$ refers to a price event on the NYSE. The associated counting functions that establish the number of s -type events over t are indexed by $N^s(t)$. Pooling and ordering of the arrival times, t_i^1 and t_i^2 , yields a simple point process $\{t_i\}_{i=1}^n$ with counting function $N(t)$. We assume that the arrival times are strictly distinct, $0 < t_1 < t_2 < \dots < t_n$. Based on this assumption, the individual point processes are strictly orderly, too. Fig. 1 gives an illustration of a pooled point process $N(t)$ consisting of two individual processes $N^1(t)$ and $N^2(t)$.

The internal filtration denoted by $\mathfrak{F}_t$ consists of the complete information path of the left continuous counting process $N(t)$. The $\mathfrak{F}_t$ -intensity process that characterizes the evolution of $N^s(t)$ is then

$$\lambda^s(t; \mathfrak{F}_t) = \lim_{\Delta \rightarrow 0} \frac{1}{\Delta} \mathbb{P}[N^s(t + \Delta) - N^s(t) > 0, N^{s'}(t + \Delta) - N^{s'}(t) = 0 | \mathfrak{F}_t] \quad (1)$$

$\forall s \neq s'$, where $s' = 1, 2$. Eq. (1) gives an instantaneous probability of observing an s -type price event conditional on the information set available at t .

Russell's (1999) ACI model defines the s -type conditional intensity function as

$$\lambda^s(t; \mathfrak{F}_t) = \lambda_0^s \psi^s(t) \phi^s(t). \quad (2)$$

The baseline intensity function that defines the deterministic evolution of the intensity function of the price events in the s^{th} market is

² We define informative price events as cumulated absolute midquote changes. A detailed description follows in Section 2.2.

¹ Grammig and Peter (forthcoming) propose a methodology to uniquely determine Hasbrouck information shares. They use distributional assumptions to identify unique information shares using daily data. Generally, the assumptions on which they base their identification, are rather suited for middle and low frequency data and often are not supported by the high frequency data, such as the data used in our empirical application.

denoted by λ_o^s . We ensure positivity of the conditional intensity function in (2) by estimating λ_o^s as an exponential function of the unknown parameter ω^s . $\psi^s(t)$ shifts the baseline function by capturing the dynamic structure of the conditional intensity. $\phi^s(t)$ accounts for diurnal patterns, which are common when dealing with high frequency financial data.

The 2×1 vector, $\psi_i = (\psi_i^1, \psi_i^2)'$, is parametrized in terms of a vector autoregressive moving average (VARMA (1,1)) process,

$$\tilde{\psi}_i = \sum_{s=1}^2 (\mathbf{a}^s \varepsilon_{i-1}^s + \mathbf{B} \tilde{\psi}_{i-1}) y_{i-1}^s, \quad (3)$$

where $\mathbf{a}^s$ is a 2×1 coefficient vector and $\mathbf{B}$ is a 2×2 autoregressive coefficient matrix. In order to ensure positivity of $\lambda^s(t; \mathfrak{T}_t)$ in (2), we let $\psi^s(t) = \exp(\tilde{\psi}_{N(t)}^s)$. y_i^s denotes an indicator variable that takes on the value one if the i^{th} price event of the pooled process comes from the TSX process ($s=1$) or the NYSE process ($s=2$), and zero otherwise. Henceforth, we label an innovation originating in the home market (TSX) with a superscript 1 and the corresponding coefficient vector with $\mathbf{a}^1$. The first element of $\mathbf{a}^1$ (a_1^1) then measures the impact of a TSX innovation on the TSX conditional intensity. The second element (a_2^1) gives the cross effect of an innovation in the TSX process on the NYSE conditional intensity. Correspondingly, we label NYSE-associated innovations with a superscript 2. We allow the autoregressive coefficient matrix $\mathbf{B}$ to be diagonal, only. The diagonal elements of $\mathbf{B}$ determine the long-run impact of an innovation and stationarity of the process is ensured if the eigenvalues of $\mathbf{B}$ lie inside the unit circle. By restricting the $\mathbf{B}$ matrix to be diagonal we follow earlier work of Russell (1999), Bauwens and Hautsch (2006) and Hall and Hautsch (2006). Additionally, we conduct in our empirical analyses goodness-of-fit comparisons between the ACI model with restricted and unrestricted $\mathbf{B}$ matrix. The results confirm that the additional two parameters in the $\mathbf{B}$ do not significantly improve the models goodness-of-fit in terms of the SIC (see Schwarz, 1978) indicating that a parsimonious specification is more suitable in order to avoid overfitting.³

According to Russell (1999), the specification of the innovation in (3) is based on the integrated intensity, which is computed by piecewise integration of $\lambda^s(t; \mathfrak{T}_t)$,

$$A^s(t_{i-1}^s, t_i^s) = \int_{t_{i-1}^s}^{t_i^s} \lambda^s(u; \mathfrak{T}_u) du = \sum_j \int_{\tilde{t}_j}^{\tilde{t}_{j+1}} \lambda^s(u; \mathfrak{T}_u) du, \quad (4)$$

with j denoting all points where $t_{i-1}^s < \tilde{t}_j < \tilde{t}_{j+1} \leq t_i^s$.

Using the random time change theorem, any non-Poisson process can be transformed into a standard Poisson process that implies an integrated intensity with iid standard exponential distribution, i.e. $A^s(t_{i-1}^s, t_i^s) \sim \text{iid Exp}(1)$ (see Hautsch, 2003; Brémaud, 1981; Bowsher, 2007). Following Bauwens and Hautsch (2006), we then define the innovation in (3) as the logarithm of an iid exponential variate centered by its unconditional expectation,⁴

$$\varepsilon_i^s = -0.5772 - \ln A^s(t_{i-1}^s, t_i^s). \quad (5)$$

Hence, an innovation in the ACI model can be interpreted as the difference between the realized number of events and the expected number of events within the interval $(t_{i-1}^s, t_i^s]$. This implies that positive values of ε_i indicate an underprediction and negative values indicate an overprediction of arrival rates.

The model parameters are estimated using the method of maximum likelihood. The log-likelihood function of the two-dimensional ACI process can be expressed as

$$\ln \mathcal{L}(\theta) = \sum_{s=1}^2 \sum_{i=1}^n \{-A^s(t_{i-1}, t_i) + y_i^s \ln \lambda^s(t_i; \mathfrak{T}_i)\}, \quad (6)$$

with $\theta = (\omega^s, \mathbf{a}^s, \mathbf{B})$.

The first term on the right-hand side of (6) corresponds to the s -type intensity integrated over $(t_{i-1}, t_i]$, and the second to the arrival time probability in the pooled process. The log-likelihood can be maximized by standard nonlinear optimization algorithms.

If the model is specified correctly, the resulting s -type residuals, $\tilde{\varepsilon}_i^s = A^s(t_{i-1}^s, t_i^s)$, should be iid unit exponentially distributed. Hence, the dynamic and distributional properties of the estimated residuals can be evaluated using the overdispersion test suggested by Engle and Russell (1998). Their test statistic against excess dispersion, $\sqrt{\frac{n^s}{8}}(\sigma_{\tilde{\varepsilon}^s}^2 - 1)$, asymptotically follows a normal distribution, where $\sigma_{\tilde{\varepsilon}^s}^2$ denotes the variance of the s -type residual series and n^s denotes the number of price events in the process s . The degree of autocorrelation in the residuals not explained by the specified model is assessed by a Ljung-Box test. However, if the assumption of exponentially distributed innovations is not supported by the data, we achieve consistent quasi-maximum likelihood estimators of θ by specifying only the conditional mean of the innovation distribution (see White, 1982; Bollerslev and Wooldridge, 1992).

2.2. The data

We examine Canadian stocks traded on the TSX and cross-listed on the NYSE. These stocks are particularly well suited for price discovery analyses, since the trading periods of the TSX and NYSE are identical. Moreover, the large number of Canadian NYSE-listed stocks allows for further cross-sectional analyses concerning the determinants of information shares. We use quote data for 83 Canadian NYSE-listed stocks. The NYSE data are extracted from the Trade and Quote (TAQ) DVDs supplied by the NYSE. Toronto quote and trade data were taken from the Equity Trades and Quotes data set provided by the TSX. The sample period covers 62 trading days from January 1, 2004 to March 31, 2004. Continuous trading on both exchanges takes place from 9:30am to 4:00pm. Table 1 gives the stock ticker and company names.

Table 1

The table shows the ticker symbols of the 83 Canadian sample stocks together with the full company name and their industries.

Ticker	Company name	Industry
ABX	Barrick Gold	Gold Mining
ABY	Abitibi Consolidated Inc.	Paper
AEM	Agnico Eagle Mines Ltd.	Gold Mining
AGU	Agrium Inc.	Chemicals (Specialty)
AL	Alcan Inc.	Metals and Mining
BCE	BCE Inc.	Foreign Telecom.
BCM	Canadian Imp. Bank of Commerce	Bank
BEI	Boardwalk Equities	Real Estate Holding
BGM	General Motors Corp.	Automobile
BMO	Bank of Montreal	Bank
BNN	Brascan Corp.	Real Estate Holding
BNS	Bank of Nova Scotia	Bank
BPO	Brookfield Properties Corporation	Real Estate Holding
BR	Broadridge Inc.	IT Services and Consulting
BVF	Biovail Corp.	Pharmaceuticals
CCJ	Cameco Corp.	Nonferrous Metals
CGT	CAE Inc.	Aerospace
CJR	Chorus Entertainment Inc.	Broadcasting and Entertainment
CLS	Celestica Inc.	Electronics
CNI	Canadian National Railway	Transport
CNQ	Canadian Natural Resources	Petroleum (Production)

(continued on next page)

³ Results on the SIC comparisons are available on request.

⁴ As noted by Hautsch (2003), the logarithm of an iid exponential variate yields a minimum Gumbel variate, $\ln A^s(t_{i-1}^s, t_i^s)$, with mean -0.5772 and variance $\sigma_{\varepsilon}^2 = \pi^2/6$.

Table 1 (continued)

Ticker	Company name	Industry
COT	Cott Corp.	Soft Drinks
CP	Canadian Pacific Railway	Transport
CWG	CanWest Global Communications Corp.	Broadcasting and Entertainment
DTC	Domtar Corp.	Paper
ECA	EnCana Corp.	Energy
ENB	Enbridge Inc.	Gas Distribution
ERF	Enerplus Resource Fund	Exploration and Production
EXEA	Extendicare Inc.	Health Services
FDG	Fording Canadian Coal Trust	Mining (Other Mines)
FFH	Fairfax Financial Holdings Ltd.	Property and Casualty Insurance
FHR	Fairmont Hotels Resorts Inc.	Hotels
FS	Four Seasons Hotels Inc.	Hotels
GG	Goldcorp Inc.	Gold Mining
GIB	CGI Group Inc.	Computer Services
GIL	Gildan Activewear Inc.	Clothing and Accessories
GLG	Glamis Golds Ltd.	Gold Mining
HBG	Hub International Ltd.	Insurance
IDR	Intrawest Corp.	Hotels
IPS	IPSCO Inc.	Metals and Mining
IQW	Quebecor World	Publishing
ITN	Intertan Inc.	Electronics
ITP	Intertape Polymer Group Inc.	Containers and Packaging
KFS	Kingsway Financial Services Inc.	Insurance
KGC	Kinross Gold Corp.	Gold Mining
LAF	Lafarge North America Inc.	Construction Materials
MDG	Meridian Gold Inc.	Gold Mining
MDZ	MDS Inc.	Medical Equipment
MFC	Manulife Financial Corp.	Insurance
MGA	Magna International Inc.	Auto Parts
MHM	Masonite International Corp.	Building Products
MIM	MI Developments Inc.	Gambling
MWI	Moore Wallace	Computer Services
N	Inco Ltd.	Metals and Mining
NCX	Nova Chemicals Corp.	Commodity Chemicals
NRD	Noranda Inc.	Metals and Mining
NT	Nortel Networks	Foreign Telecom.
NDY	Nexen Inc.	Energy
OPY	Oppenheimer Holdings Inc.	Investment Services
PCZ	Petro-Canadian Com.	Integrated Oil and Gas
PDG	Placer Dome	Precious Metals
PDS	Precision Drilling Corp	Oil Equipment and Services
PGH	Pengrowth Energy	Exploration and Production
PKZ	PetroKazakhstan Inc.	Petroleum
POT	Potash Corp.	Chemicals
PWI	Primewest Energy Trust	Energy
RBA	Ritchie Bros Auctioneers	Industrial Equipment
RCN	Radiant Communications	Telecommunications
RG	Rogers Publishing Limited	Publishing
RY	Royal Bank of Canada	Bank
RYG	Royal Group Technologies Ltd.	Building Products
SLF	Sun Life Financial Serv.	Insurance
SU	Suncor Energy	Petroleum
TAC	TransAlta Corp.	Conventional Electricity
TD	Toronto-Dominion	Bank
TEU	CP Ships Ltd.	Maritime
TLM	Talisman Energy	Energy
TOC	Thomson Corp.	Information Services
TRA	Terra Industries	Chemicals
TRP	TransCanada Corp.	Energy
TU	Telus Corp.	Telecommunications
VTG	Veritas DGC Inc.	Energy
ZL	Zarlink Semiconductor Inc.	Semiconductors

Table 2

Descriptive statistics across 83 sample stocks.

Variable	Q5	Q25	Mean	Median	Q75	Q95
Panel A						
<i>Spread</i> ¹	0.020	0.034	0.082	0.052	0.093	0.266
<i>Spread</i> ²	0.023	0.033	0.061	0.048	0.077	0.135
<i>Spread</i>	0.61	0.87	1.93	1.17	1.55	5.38
<i>MedTrad</i>	0.60	0.87	1.47	1.11	1.82	3.28
<i>TsxVol</i>	0.01	0.45	0.64	0.73	0.92	0.98
Panel B						
#Q ¹	28.8	74.6	158.3	111.2	208.5	428.8
#Q ²	38.7	92.4	180.7	153.3	247.2	406
τ^1	00:52	01:51	04:23	03:20	04:57	12:33
τ^2	00:57	01:33	03:25	02:28	04:04	08:55

The table displays the five percent quantile (Q5), the first quartile (Q25), the mean (Mean), the Median (Median), the third quartile (Q75), and 95 percent quantile (Q95) across 83 sample stocks. For the TSX the superscript $s = 1$ and for the NYSE $s = 2$. *Spread*¹ is the spread on the TSX in Canadian Dollars and *Spread*² is the spread on the NYSE in US Dollars. *Spread* denotes the ratio of the percentage quoted spread on the TSX to the percentage quoted spread on the NYSE. *MedTrad* is the number of the medium shares to total shares traded on the TSX relative to the ratio of the medium shares to total shares traded on the NYSE. A medium trade is defined in medium-sized lots of 2501–10,000 shares. *TsxVol* is given by the number of shares traded on the TSX relative to the total number of shares traded on the TSX and the NYSE. The rows labeled #Q^s give the average number of observations per day. The rows labeled τ^s contain the average transaction durations in mm:ss. All statistics are calculated over the sample period from January 1 to March 31, 2004.

first introduced a thinning rule to the literature and choose a rather small value of 0.0005 US\$ which in their sample corresponds to half the largest observed spread. Bauwens and Hautsch (2006) selected threshold values between 0.08 US\$ and 0.46 US\$ using multiples of 20 of the mean trade and quote changes. In our empirical analysis the mean spread on the TSX is 0.08 CAD and on the NYSE 0.06 US\$. We set threshold values to 0.025 Canadian Dollars for the TSX, which, after accounting for the exchange rate and the minimum tick size, corresponds to 0.02 US Dollars for the NYSE price changes. Our threshold corresponds roughly to one third of the mean spread and half of the median spread on the two markets (compare Panel A of Table 2). This selection rule is similar to Bowsher (2007) whose threshold is given by half the mean spread. Since we conducted robustness checks for our empirical analysis using the next higher (TSX: 0.03 CAD and NYSE: 0.025 US\$) and next lower (TSX: 0.02 CAD and NYSE: 0.015 US\$) thresholds, we are confident that our results and conclusions do not heavily depend on this rather arbitrary selection rule.⁵

The thinning algorithm is applied to reduce the amount of “noise” due to microstructure effects that should not be considered as a change in the fundamental price. As a result, we obtain what we refer to as informative price events. The ACI model introduced in the previous section assigns zero probability to the simultaneous occurrence of two events, and therefore quote revisions with the same time stamp within one market are treated as one. Finally, events with the same time stamp in both markets are deleted.⁶

Additionally, Table 2 presents summary statistics for the liquidity proxies used in the cross-sectional analysis. The liquidity-related variables are the relative spread (*Spread*), measured by the percentage quoted spread on the TSX relative to the percentage quoted spread on the NYSE, and traded volume (*TsxVol*), given by the number of shares traded on the TSX relative to the total number of shares traded on the TSX and the NYSE combined.

Hasbrouck (1995) classifies trades into volume categories and finds that a relatively higher number of medium-sized trades is positively correlated with the relative information share. We compute medium trades *MedTrad* as the number of the medium shares

⁵ Results are available upon request.

⁶ On average, across all sample stocks only 5% of the data are affected by this selection rule.

Prior to estimation, the data have to be thinned in order to reduce the amount of noise and retrieve a time series of price-relevant events. Following Engle and Russell (1997), Bowsher (2007) and Bauwens and Hautsch (2006), the quote data are thinned using a function of price marks. Bid and ask midquotes are calculated and absolute cumulative price changes constructed that are retained whenever they exceeded a specific threshold. Generally, finance literature does not provide explicit guidance concerning the thinning of data for point process modeling. Engle and Russell (1997)

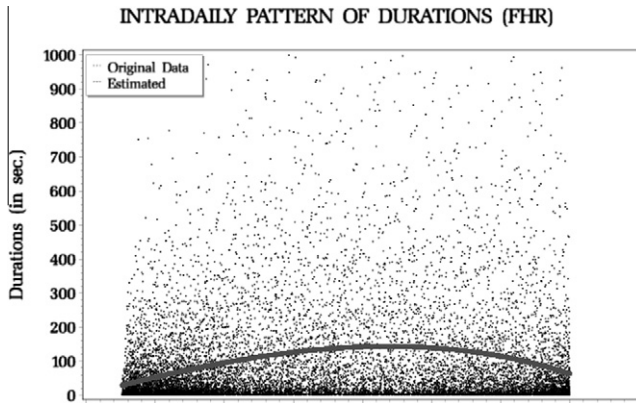


Fig. 2. Intraday pattern of durations. The figure shows the transaction durations (original data: black dots) of the pooled process for a randomly chosen stock (FHR). As visible from the figure, transaction durations exhibit a $\cap$ -shape. The typical intraday pattern is captured by the estimated seasonal figure (estimated: line).

to total shares traded on the TSX relative to the ratio of the medium shares to total shares traded on the NYSE.⁷

Table 2 in Panel B gives summary statistics for the filtered data. Across 83 stocks, the average daily number of quote revisions is 158 on the TSX and 181 on the NYSE. Quote revisions on the TSX and NYSE occur on average every 4.5 min and 3.5 min, respectively. On the TSX, the most frequent stock has a midquote change every 23 s, and the least quoted stock updates its midquotes on average every 21 min. Accordingly, the NYSE duration averages range from 46 s to 20 min.

Several authors (e.g. Engle and Russell, 1997) observe that price durations exhibit an intraday pattern in the rate of arrival. To account for this, we diurnally adjust the data prior to estimation. Assuming the separability of the time function and stochastic function in (2), the elimination of the time-of-day effects proceeds in two steps. First, the typical intraday pattern (ϕ_i) is estimated by regressing the transaction durations ($\tau_i = t_i - t_{i-1}$) of the pooled process on polynomial and trigonometric time functions (see Eubank and Speckman, 1990 and Appendix A). Second, dividing the durations by their estimated typical shape gives intraday adjusted durations, i.e. $\tilde{\tau}_i = \frac{\tau_i}{\phi_i}$. Finally, a diurnally adjusted arrival times series in the pooled process is achieved by setting the first arrival time of the day to zero and cumulating adjusted durations for each day.

Fig. 2 shows the quote durations for one of our sample stocks before removing the intraday pattern. As is clearly visible from the figure, transaction durations exhibit the typical $\cap$ -shape. The intraday pattern is captured by the estimated intraday effects.

2.3. Intensity-based information shares

In order to measure contributions to price discovery, we draw our attention to the cross effects of the conditional intensities associated with quote changes on the home and foreign markets. As prices in both markets refer to the same underlying asset, the law of one price holds. This implies that informative price events occurring in one market will subsequently be incorporated into the price of the other market to offset arbitrage opportunities. Consequently, unexpected quote adjustments in one market trigger price events in the other market and increase the other market's conditional intensity. Analogously, the unexpected absence of price changes in one market reduces the other market's intensity. These interactions entail positive cross effects. We propose using the size of the markets' cross effects to measure the relative importance of a market in the price discovery process.

⁷ As in Hasbrouck (1995) and Eun and Sabherwal (2003) a medium trade is based on a trade category of 2501–10,000 shares.

We start by examining the long-run impacts of an innovation on the conditional intensities by analyzing the full dynamics of the markets' interdependencies.

For this purpose, we compute impulse response functions and derive the cumulated effects of an innovation in period i on $\tilde{\psi}_{i+h}$. Iterating the ARMA specification in (3) h periods forward yields

$$\begin{aligned} \tilde{\psi}_{i+h} = & \sum_{s=1}^2 \mathbf{a}^s \varepsilon_{i+h-1}^s y_{i+h-1}^s + \mathbf{B} \sum_{s=1}^2 \mathbf{a}^s \varepsilon_{i+h-2}^s y_{i+h-2}^s + \dots \\ & + \mathbf{B}^{h-2} \sum_{s=1}^2 \mathbf{a}^s \varepsilon_{i+1}^s y_{i+1}^s + \mathbf{B}^{h-1} \sum_{s=1}^2 \mathbf{a}^s \varepsilon_i^s y_i^s + \mathbf{B}^h \tilde{\psi}_i. \end{aligned}$$

To isolate the standard deviation innovation $\sigma_\varepsilon = \sqrt{\pi^2/6}$ in period i , all subsequent innovations are set to their unconditional mean, $\mathbb{E}[\varepsilon_i^s] = 0$. Further, the unconditional mean $\mathbb{E}[\tilde{\psi}_i] = 0$ is used as a starting value for $\tilde{\psi}_i$. The impulse response functions are then given by,

$$IR^1(h) = \mathbf{B}^{h-1} \mathbf{a}^1 \sigma_\varepsilon \quad \text{and} \quad IR^2(h) = \mathbf{B}^{h-1} \mathbf{a}^2 \sigma_\varepsilon, \quad (7)$$

where IR^1 denotes the bivariate impulse response function associated with an innovation in the TSX price process, and IR^2 correspondingly gives the bivariate impulse response function associated with an innovation in the NYSE price process. Totalling the effects in each period delivers the cumulative impulse response functions,

$$CIR^1(h) = \sum_{j=1}^h \mathbf{B}^{j-1} \mathbf{a}^1 \sigma_\varepsilon \quad \text{and} \quad CIR^2(h) = \sum_{j=1}^h \mathbf{B}^{j-1} \mathbf{a}^2 \sigma_\varepsilon. \quad (8)$$

For a stationary process, the effects of an innovation die out in the long-run. Thus, the cumulative impulse response functions in (8) for $h \rightarrow \infty$ converge to the finite vectors given below:

$$\lim_{h \rightarrow \infty} CIR^1 = [1 - \mathbf{B}]^{-1} \mathbf{a}^1 \sigma_\varepsilon \quad \text{and} \quad \lim_{h \rightarrow \infty} CIR^2 = [1 - \mathbf{B}]^{-1} \mathbf{a}^2 \sigma_\varepsilon. \quad (9)$$

Fig. 3 shows cumulated impulse response functions as derived in (8) averaged over the sample stocks. The left panel depicts the impact of a standard deviation innovation in the TSX intensities and its impacts on TSX and NYSE processes. Correspondingly, the right panel illustrates the impacts of a NYSE standard deviation innovation.

Corresponding to the arbitrage relation between prices in both markets, we find positive spillover effects, i.e. positive intensity innovations have significant positive impacts on the other market's conditional intensity.

More precisely, the other market experiences an increase in the frequency of quote adjustments, associated with an information flow into this market. The extreme case of no cumulated cross impacts would indicate that the market's conditional intensity does not react to unexpected events in the other market. As expected, own market innovations, i.e. TSX $\rightarrow$ TSX and NYSE $\rightarrow$ NYSE, have larger impacts than innovations in the other market, i.e. TSX $\rightarrow$ NYSE and NYSE $\rightarrow$ TSX. The long-run effects of a TSX innovation on the NYSE intensity tend to be larger than the effects of a NYSE innovation on the TSX intensity. These differences in the markets' reactions might be due to differences in the market characteristics of the TSX and NYSE, and should be considered when comparing effects across markets.

Hence, in order to determine the contributions to price discovery, we focus on the magnitude of cumulated cross effects in either market (CIR_2^1 and CIR_1^2) and account for differences in the markets' specific adjustment to intensity innovations. We therefore suggest *standardizing* the cumulative cross effects using the cumulative impacts of the own market innovation. Consequently, $\frac{CIR_2^2}{CIR_1^1}$ denotes the cross effects of a NYSE intensity innovation on the TSX conditional

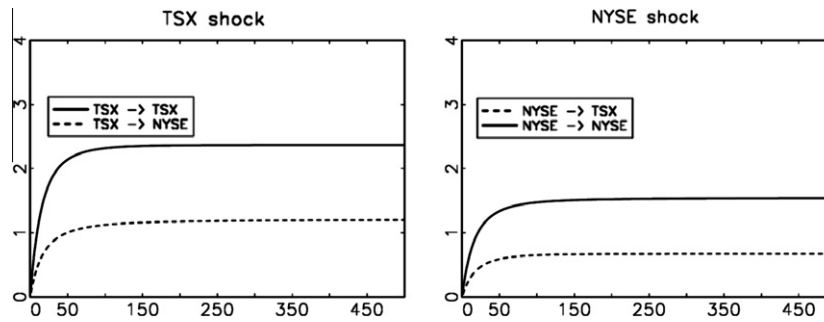


Fig. 3. Cumulated impulse response function of a standard deviation shock. The figure shows cumulated impulse response functions of (8) for the recursive process $\tilde{\psi}$ in (3) averaged across 73 sample stocks (see Table 1). The left panel depicts the impacts of a standard deviation shock on the TSX and its impacts on the TSX (solid line) and NYSE (dashed line) processes. Analogously, the right panel illustrates the impact of a NYSE standard deviation shock on the NYSE (solid line) and TSX (dashed line) processes respectively.

Table 3

Estimation summary results.

Estimates	Mean	Std	Q25	Q75	Min	Max	M(SE)	#sig
$\hat{\omega}^1$	-1.285	0.329	-1.499	-1.079	-2.031	-0.169	0.0501	73
$\hat{\omega}^2$	-0.936	0.286	-1.150	-0.737	-1.630	-0.278	0.0542	73
$\hat{a}_1^1$	0.116	0.030	0.095	0.131	0.051	0.213	0.0147	73
$\hat{a}_2^1$	0.052	0.023	0.039	0.062	0.005	0.171	0.0096	70
$\hat{a}_1^2$	0.030	0.017	0.020	0.035	0.004	0.132	0.0069	70
$\hat{a}_2^2$	0.070	0.031	0.046	0.088	0.020	0.178	0.0109	73
$\hat{b}^1$	0.927	0.032	0.905	0.947	0.834	0.981	0.0147	73
$\hat{b}^2$	0.929	0.038	0.905	0.955	0.801	0.993	0.0186	73

The table contains descriptive statistics for the estimated parameters of the ACI model in (2) and (3) computed over 73 sample stocks that have positive $\hat{a}^s$ estimates. The table displays the mean (*Mean*), the standard deviation (*Std*), the first quartile (*Q25*) and third quartile (*Q75*), and the minimum (*Min*) and maximum (*Max*) of the estimated parameters. *M(SE)* is the average standard error of the estimates and *#sig* gives the number of significant estimates at the 5% significance level.

intensity, standardized using the impacts of a TSX innovation on TSX's intensity. $\frac{CIR_1^1}{CIR_2^2}$ gives the analog ratio for the NYSE. If we consider the Equations in (9), we find that this ratio cancels down to $\frac{a_1^1}{a_2^1}$ for the TSX and $\frac{a_2^2}{a_1^2}$ for the NYSE. Finally, in order to simplify the interpretation of our measure, we confine the information shares to lie between zero and one by taking the standardized cross effects of each market relative to the sum of the standardized cross effects:

$$IIS^1 = \frac{\frac{a_1^1}{a_2^1}}{\frac{a_1^1}{a_2^1} + \frac{a_2^2}{a_1^2}} \quad \text{and} \quad IIS^2 = \frac{\frac{a_2^2}{a_1^2}}{\frac{a_1^1}{a_2^1} + \frac{a_2^2}{a_1^2}}. \quad (10)$$

Eq. (10) gives the unique intensity-based information shares, where IIS^1 denotes the TSX contribution to price discovery and IIS^2 that of the NYSE. Standard errors of the IIS in (10) can be computed via the delta method.

3. Estimation results and discussion

As outlined in Section 2.1, estimation of the model parameters is done via maximizing the likelihood function in (6). We do not allow previous-day innovations to affect the next day's intensity; hence, the likelihood function is re-initialized each day and becomes the sum of independent day-likelihoods.

Table 3 contains descriptive statistics for the estimated ACI coefficients averaged over 73 stocks which have positive $\hat{a}^s$ estimates.⁸ Stock-specific results can be found in Table 7 in Appendix B.

The estimated baseline intensities exhibit small standard errors and are statistically significant at the 1% level of confidence. The cross-sectional correlation between the ratio of baseline coefficients, $\frac{\exp(\hat{\omega}^1)}{\exp(\hat{\omega}^1) + \exp(\hat{\omega}^2)}$, and the number of quotes on the TSX relative to the total number of quotes on the TSX and the NYSE is 0.9. Therefore, the baseline function captures the differences in the intensity levels in the two markets extremely well.

According to Table 3, the short-run impacts of TSX innovations on the NYSE process are on average about twice as large as the effects of NYSE innovations on the TSX intensity. For 70 stocks, the innovation cross effects are significant at the 5% level. As we would expect from the duration modeling literature (e.g. Engle and Russell, 1998), we find strong persistence of innovations, which is reflected in the large autoregressive coefficients. Across our sample stocks, $\hat{b}^1$ is on average 0.927 and $\hat{b}^2$ is 0.929.

Turning to the test statistics assessing the model specification, we find that the results for the Ljung-Box test in Table 4 of the estimated residuals are mixed. For some stocks, the null hypothesis of no autocorrelation cannot be rejected. Further, the table shows that the means of the estimated residuals are on average close to one for both markets. Overall, the results indicate that some excess dispersion is still present, nevertheless we obtain consistent estimators based on the quasi-maximum likelihood method.

Table 5 displays summary results for the unique intensity-based information share according to (10). Stock-specific results can be found in Table 8 in Appendix B. As the Hasbrouck methodology is a well-established measure, we also report Hasbrouck information shares. Additionally, we present results for the adjustment coefficient ratios.⁹

⁸ The stocks which are excluded according to these criteria are BEI, BR, CJR, FFH, and OPY. BGM and ITN were excluded prior to estimation due to erroneous data. VTS, XEA, and LAF do not achieve proper convergence.

⁹ Details of deriving Hasbrouck information shares and adjustment coefficient ratios are outlined in Appendix C.

Table 4
Residual diagnostics for the ACI model.

Ticker	TSX					NYSE				
	$\bar{\varepsilon}^1$	σ_{ε^1}	OD^1	AC^1	LB^1	$\bar{\varepsilon}^2$	σ_{ε^2}	OD^2	AC^2	LB^2
ABX	1.00	1.32	26.93	0.05	22.14	1.00	1.17	14.41	0.04	20.40
ABY	1.04	1.46	16.85	0.02	0.87	1.02	1.43	18.97	0.03	2.11
AEM	1.00	1.40	27.84	0.01	0.67	1.02	1.35	26.10	0.03	6.32
AGU	1.01	1.62	41.36	0.02	2.52	1.01	1.44	28.01	0.03	6.31
AL	1.00	1.38	51.42	0.02	6.85	1.00	1.30	43.45	0.04	45.18
BCE	1.00	1.41	21.11	0.03	2.98	1.01	1.37	23.72	0.03	5.22
BCM	0.99	1.50	48.43	0.01	2.09	1.01	1.54	70.42	0.03	25.14
BMO	1.00	1.50	42.89	0.04	18.30	1.00	1.54	63.86	0.03	12.29
BNN	1.01	1.73	61.05	0.01	0.72	1.01	1.75	85.60	0.05	41.15
BNS	0.99	1.52	39.97	0.01	0.38	1.01	1.56	63.06	0.04	26.77
BPO	1.03	1.92	63.40	0.04	6.59	1.00	1.58	48.88	0.05	18.12
BVF	1.01	1.54	60.50	0.06	55.63	1.01	1.45	47.39	0.07	80.92
CCJ	0.99	1.77	92.18	0.05	32.65	1.01	1.73	113.79	0.05	76.00
CGT	1.05	1.50	11.77	−0.02	0.29	1.04	1.39	10.25	0.08	5.91
CLS	1.01	1.46	44.92	0.06	50.52	1.00	1.44	42.75	0.06	44.53
CNI	0.99	1.62	71.95	0.04	23.46	1.00	1.47	56.05	0.02	8.44
CNQ	0.98	1.54	59.77	0.03	10.66	1.00	1.65	106.15	0.02	14.03
COT	1.01	1.77	57.50	−0.01	0.47	1.02	1.60	50.79	0.03	7.22
CP	1.00	1.72	57.51	−0.01	1.39	1.01	1.42	36.68	0.05	25.18
CWVG	1.01	2.27	83.99	0.00	0.06	1.03	1.90	47.16	0.10	24.96
DTC	1.02	1.59	28.59	0.00	0.01	1.03	1.42	22.92	0.03	2.68
ECA	0.99	1.46	49.55	0.03	16.26	1.01	1.30	32.17	0.06	59.40
ENB	1.06	2.07	79.71	0.01	0.17	1.01	1.75	84.97	0.00	0.24
ERF	1.01	1.70	62.99	0.06	30.93	1.02	1.55	53.21	0.06	36.68
FDG	1.01	1.92	112.31	0.07	62.91	1.01	2.01	151.04	0.07	103.29
FHR	1.02	1.71	56.17	0.02	3.43	1.00	1.69	57.72	0.01	0.99
FS	1.01	1.66	80.52	0.06	57.52	1.00	1.74	92.32	0.03	12.16
GG	1.02	1.37	27.16	0.05	16.20	1.01	1.21	15.37	0.08	61.91
GIB	1.04	1.53	15.16	−0.03	1.04	1.02	1.56	20.09	0.04	2.37
GIL	1.02	2.14	92.95	0.04	8.83	1.01	2.13	130.89	0.05	25.49
GLG	1.01	1.52	43.99	0.04	12.29	1.01	1.37	31.83	0.06	39.80
HBG	1.03	1.94	81.02	0.06	21.74	1.01	1.93	61.17	0.08	28.71
IDR	1.01	1.78	56.06	0.03	4.83	1.01	1.66	50.42	0.02	2.39
IPS	1.03	1.97	59.37	0.03	2.30	1.02	1.90	67.38	0.07	27.57
IQW	1.02	1.82	55.23	0.01	0.48	1.00	1.55	42.79	0.07	33.52
ITP	1.03	1.91	52.96	0.04	4.50	1.02	1.98	76.35	0.03	6.53
KFS	1.07	1.97	44.44	0.03	2.33	1.02	1.69	42.80	0.04	5.88
KGC	1.04	1.37	16.68	0.04	3.76	1.02	1.19	8.62	0.07	16.18
MDG	1.01	1.49	34.65	0.03	4.34	1.01	1.33	25.06	0.03	9.98
MDZ	1.02	1.65	33.02	0.02	0.88	1.03	1.58	34.88	0.05	10.18
MFC	1.00	1.47	36.84	0.04	10.20	1.00	1.52	48.22	0.02	6.42
MGA	0.99	1.66	89.45	0.04	28.85	1.00	1.48	64.74	0.02	10.84
MHM	1.00	2.10	83.03	0.00	0.00	1.01	1.97	108.71	0.04	17.56
MIM	1.02	1.96	82.26	0.03	6.64	1.00	1.95	83.27	0.02	4.04
MWI	1.03	1.93	70.35	0.08	32.93	1.01	1.73	43.38	0.03	4.53
N	1.00	1.42	57.20	0.04	38.40	1.00	1.40	53.39	0.05	51.21
NCX	1.02	1.65	46.38	0.03	3.69	1.00	1.49	39.98	0.02	3.52
NRD	0.99	1.48	34.65	0.02	1.76	1.00	1.48	41.62	0.03	6.41
NT	0.97	1.09	4.86	0.05	10.53	0.97	1.13	6.07	0.01	0.12
NXY	0.99	1.67	67.34	0.02	4.59	1.00	1.73	100.08	0.03	18.76
PCZ	0.99	1.60	62.38	0.04	19.34	1.02	1.58	75.86	0.04	26.52
PDG	1.01	1.30	25.25	0.04	16.42	1.00	1.21	17.58	0.03	8.90
PDS	0.99	1.86	111.20	0.02	4.82	1.00	1.55	64.08	0.04	22.69
PGH	1.06	1.92	64.56	0.07	20.22	1.05	1.69	51.12	0.07	29.07
PKZ	0.99	2.12	167.71	0.04	24.13	1.01	1.73	92.52	0.05	45.61
POT	0.99	1.86	146.13	0.05	68.52	0.99	1.67	95.75	0.04	31.37
PWI	1.03	1.59	38.14	0.03	5.82	1.03	1.49	32.48	0.08	37.45
RBA	1.01	2.21	100.67	−0.03	4.50	1.01	2.07	97.92	0.08	41.63
RCN	1.00	2.71	188.03	0.00	0.04	1.02	2.42	199.52	0.02	8.11
RG	1.00	1.84	59.01	−0.01	0.16	1.02	1.55	42.62	0.02	4.52
RY	0.99	1.46	38.76	0.02	4.46	1.00	1.43	47.91	0.04	22.19
RYG	0.99	1.95	66.46	0.03	5.55	1.02	1.76	59.05	0.07	27.84
SLF	0.99	1.53	44.03	0.03	6.99	1.01	1.39	36.96	0.03	9.01
SU	1.00	1.51	46.44	0.02	3.10	1.02	1.41	38.96	0.05	26.12
TAC	1.10	1.89	36.60	0.00	0.02	1.02	1.68	38.47	−0.01	0.43
TD	1.00	1.41	28.52	0.02	4.05	1.01	1.43	38.37	0.02	6.02
TEU	1.03	1.73	45.89	0.00	0.00	1.01	1.62	47.52	0.04	12.67
TLM	0.99	1.64	70.05	0.00	0.26	1.01	1.53	74.69	0.03	21.39
TOC	0.99	1.66	49.05	0.03	6.49	1.02	1.54	47.93	0.04	14.63
TRA	1.02	3.07	216.86	0.02	1.86	1.06	1.94	56.44	0.07	17.78
TRP	1.01	1.57	31.08	0.04	4.92	1.02	1.35	24.52	0.05	18.36
TU	1.01	1.77	58.69	0.02	2.66	1.02	1.82	92.45	0.01	2.61
ZL	1.04	1.66	22.98	0.04	2.73	1.02	1.46	15.41	0.03	1.08

The table presents residual diagnostics for the estimated residuals corresponding to the TSX ($s = 1$) and the NYSE ($s = 2$). AC^s denotes the value of the first-order autocorrelation and the columns labeled LB^s contain the corresponding Ljung-Box statistic. $\bar{\varepsilon}^s$ and σ_{ε^s} contain the mean and the standard deviation of the estimated residuals and OD^s gives the test statistic of the overdispersion test suggested by [Engle and Russell \(1998\)](#). This statistic has a limiting normal distribution under the null with a 5% critical value of 1.645. The statistics are computed for 73 stocks that have positive α^s estimates. For full company names, see [Table 1](#).

The average home market intensity-based information share (IIS) amounts to 73%, which implies clear domination of the TSX in the price discovery process. The adjustment coefficient ratios

(Adj) also support the TSX as the leading market and match the results of [Eun and Sabherwal \(2003\)](#), who examine Canadian stock data from a period in 1999. They conclude that about two thirds

Table 5
Intensity-based information shares–descriptives.

Descriptive	TSX					NYSE				
	IIS^1	HIS_{low}^1	HIS_{up}^1	HIS_{mid}^1	Adj^1	IIS^2	HIS_{low}^2	HIS_{up}^2	HIS_{mid}^2	Adj^2
Mean	72.6	39.4	86.2	62.8	67.6	27.4	13.8	60.6	37.2	32.4
$M(SE)$	6.2	2.9	1.9	2.3	2.6	6.2	1.9	2.9	2.3	2.6
Std	17.6	24.1	19.7	20.3	26.0	17.6	19.7	24.1	20.3	26.0
Q25	63.2	15.9	81.3	48.2	50.0	15.1	1.0	39.7	20.7	9.2
Q75	84.8	59.5	98.7	78.3	90.6	36.2	18.6	83.9	49.8	49.7
Min	17.1	0.0	1.1	0.6	0.2	1.3	0.0	17.6	8.8	0.0
Max	98.7	82.4	100.0	91.2	100.0	82.9	98.9	100.0	99.4	99.8

The table presents cross-sectional descriptives in percent for the intensity-based information shares (IIS^s), Hasbrouck information share midpoints together with lower and upper bounds (HIS_{mid}^s , HIS_{low}^s and HIS_{up}^s), and the adjustment coefficient ratios (Adj^s). For the TSX, $s = 1$ and for the NYSE, $s = 2$. The descriptives include the mean (*Mean*), the standard deviation (*Std*), the mean of the information share standard error ($M(SE)$), the 25% quantile (*Q25*), the 75% quantile (*Q75*), the minimum (*Min*) and maximum (*Max*). The descriptives are computed across 73 stocks that have positive a^s estimates.

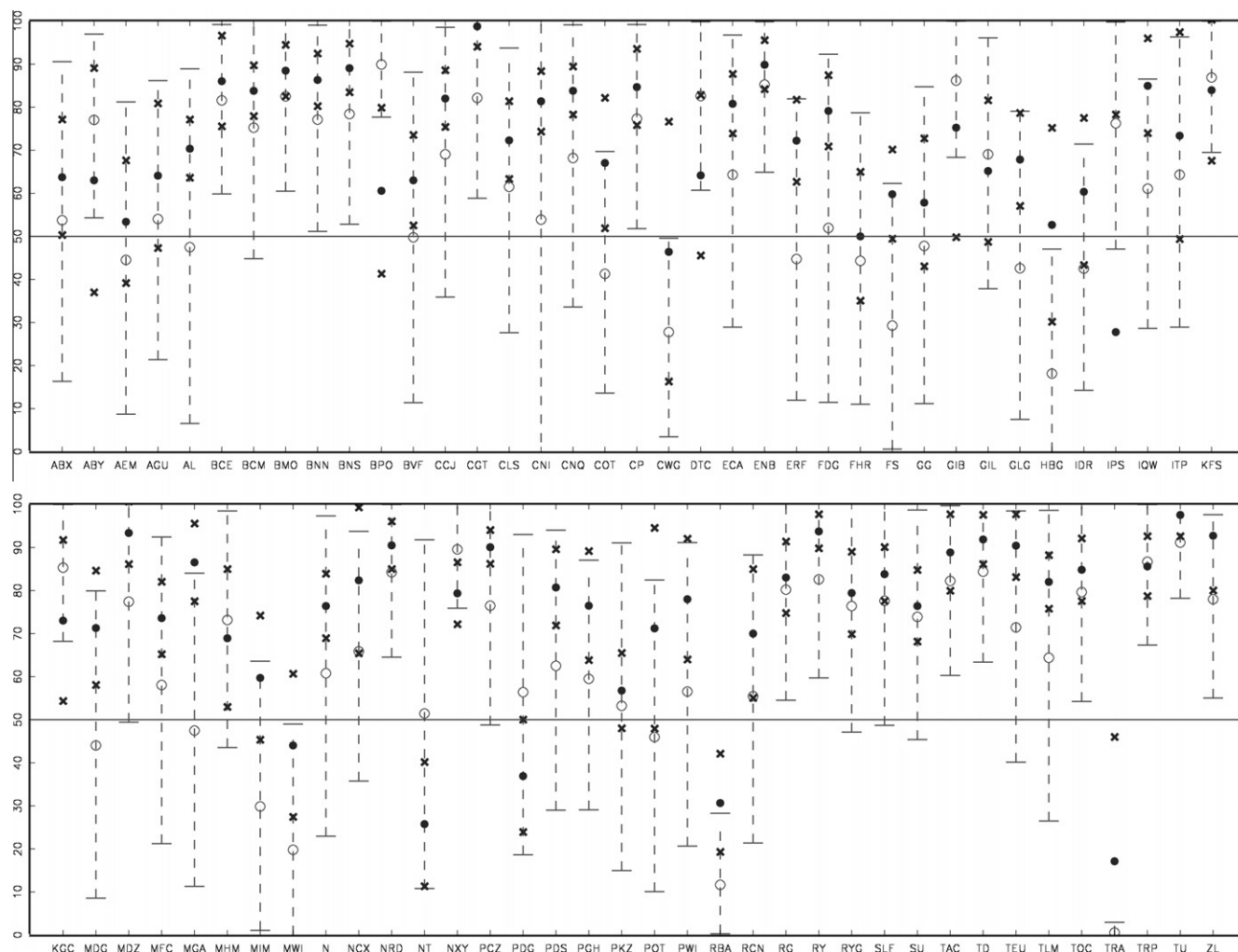


Fig. 4. Stock-specific intensity-based and Hasbrouck information shares on the TSX. The figure shows the estimated intensity-based information shares (solid circles) and their 0.95 confidence bounds (crosses), as well as the Hasbrouck information share midpoints (circles) together with the upper 0.95 confidence interval for the upper HIS share and the lower 0.95 confidence interval for the lower HIS share (vertical dashed lines).

of adjustment is done by NYSE prices. The Hasbrouck information share midpoints (HIS) deliver similar results, with an average TSX midpoint of 63%.

Stock-specific findings are illustrated in Fig. 4. The figure shows the kursiv along with their 0.95 confidence bounds, and the Hasbrouck information share midpoints with the upper 0.95 confi-

dence interval for the upper HIS share and the lower 0.95 confidence interval for the lower HIS share.

Since the HIS bounds differ on average by 47 percentage points the leading market can be determined only for 28 out of 73 stocks. For the remaining stocks, the HIS bounds do not allow for any conclusion to be drawn concerning the importance of the trading

Table 6

Cross-sectional regression results.

Explanatory variable	(1)	(2)	(3)	(4)	(5)
Constant	1.120 (1.767)	0.540 (2.300)	−1.175 (1.663)	−1.515 (1.629)	−1.323 (1.513)
LMktCap	0.574 (0.786)	−0.070 (1.133)	0.283 (0.855)	0.416 (0.855)	0.170 (0.808)
Spread	−0.925*** (0.162)				
MedTrad		0.507*** (0.139)			
TsxVol			2.551*** (0.412)	2.522*** (0.451)	2.643*** (0.459)
YearListed				−0.004 (0.003)	−0.002 (0.004)
FSale				0.002 (0.003)	0.004 (0.003)
Mining					0.126 (0.312)
Manufacturing					−0.075 (0.361)
Transport					0.370 (0.302)
Finance					0.353 (0.313)
R ² _{adj}	0.423	0.150	0.432	0.423	0.413

The table presents cross-sectional OLS estimates. The dependent variable is the logistic transformation of the TSX intensity-based information share, $\ln\left(\frac{IIS^1}{1-IIS^1}\right)$. The regression includes 73 stocks that have positive a^s estimates. *LMktCap* is the log market capitalization as reported on December 31, 2003 in the *TSX Factbook* (2003). *Spread* denotes the ratio of the percentage quoted spread on the TSX to the percentage quoted spread on the NYSE. *MedTrad* is the number of the medium shares to total shares traded on the TSX relative to the ratio of the medium shares to total shares traded on the NYSE. A medium trade is defined in medium-sized lots of 2501–10,000 shares. *TsxVol* is given by the number of shares traded on the TSX relative to the total number of shares traded on the TSX and the NYSE. *YearListed* denotes the number of years a company has been listed on the NYSE and *FSale* is the ratio of foreign sales to total sales. The remaining variables are sector dummies. We include in the regression five industry groups. Mining, Manufacturing, Finance, and Transport/Utility are dummies corresponding to four of these groups. The fifth industry group serves as benchmark sector and includes service and retail firms. R_{adj}^2 gives the regression's adjusted *R*-squared. The numbers in parentheses below the estimates are standard errors computed from the heteroskedasticity-consistent covariance matrix following White (1982).

*** Indicate the corresponding statistical two-tailed significance at $\alpha = 1\%$, respectively.

venues for the formation of the fundamental price. Using the *IIS*, we are able to answer the question “which market leads?” for 54 out of 73 stocks. We find 50 stocks for which the TSX significantly dominates price discovery, and in only 4 cases we detect a higher NYSE information share. These findings emphasize the advantage of the intensity-based information share, which is an accurate means of determining a market's contribution to the price discovery process.¹⁰

4. Cross-sectional analysis

As visible from the descriptive statistics in Table 5 and Fig. 4, the variation of the information shares across the sample stocks is considerably high. An interesting task is therefore to analyze the factors that influence a market's intensity-based information share by running cross-sectional regressions.

For this purpose, we use the TSX information share as the dependent variable and perform a logistic transformation, $\ln\left(\frac{IIS^1}{1-IIS^1}\right)$, to ensure that predicted values lie within zero and one. We follow previous studies by Korczak and Phylaktis (2010), Eun and Sabherwal (2003) and Grammig et al. (2005) and select a set of liquidity-related variables and firm-specific factors.

The liquidity-related variables used in the regressions are the relative spread (*Spread*), the traded volume (*TsxVol*) and medium trades (*MedTrad*).

We also control for firm-and sector-specific properties and use sector dummies as explanatory variables. Furthermore, we include

the number of years a firm has been listed on the NYSE up until 2004 (*YearListed*) and *FSale* as the ratio of foreign sales to total sales. Additionally, the regression contains the firm size, measured by the log of a firm's market capitalization, *LMktCap*.¹¹

As *Spread*, *MedTrad* and *TsxVol* are highly correlated, we introduce them into the regressions separately. Thus, regressions (1)–(3) in Table 6 control for the firm size and each contains a proxy for liquidity. Concerning the *Spread*, the results reveal a negative and statistically significant coefficient at the 1% level, which implies that a lower relative spread on the home market is linked to a higher TSX information share. The coefficient of the relative proportion of medium trades (*MedTrad*) and the coefficient for *TsxVol* are positive and significant at the 1% level. As regressions (4) and (5) show, the firm-specific variables are not statistically significant. Neither we find evidence for a relation between a firm's size and a market's information share, nor do the duration of listing on the NYSE, the proportion of a company's foreign sales or industry characteristics have any significant effects.

Using the *IIS*, we are able to confirm a positive link between the relative liquidity of a market and its contribution to price discovery (e.g. Eun and Sabherwal, 2003; Korczak and Phylaktis, 2010).

As acknowledged by Grammig et al. (2005), these findings indicate a link between those two variables rather than a causal relation. Nevertheless, we argue that liquidity-related variables rather than stock-specific factors determine a market's dominance, and thus suggest that market design matters for price discovery. This implies that providing an efficient trading environment and attracting investors in order to increase liquidity is of special interest for small national stock exchanges that seek to maintain their dominance in the price discovery process of cross-listed stocks.

¹⁰ Concerning the adjustment coefficient ratio, we can determine the TSX as the leading market for 49 stocks, and the NYSE leads in 14 cases. However, we focus on the comparison between the intensity-based and the Hasbrouck information shares, since the adjustment coefficient ratio as a price discovery measure has been heavily criticized (compare Lehmann, 2002; Hasbrouck, 2002; De Jong, 2002).

¹¹ Market capitalization as reported on 31 December 2003 in the *TSX Factbook* (2003).

5. Conclusion

The decisions made by investors and companies, respectively, regarding whether to invest or list their stocks depends on an exchange's ability to provide a prospering trading environment. As a result of increasing globalization and improved technology, small national exchanges are beginning to fear that they will lose their attractiveness for investors and companies. In particular, within the context of international cross-listed stocks, it is of paramount concern for a national stock exchange to remain the dominant market with regard to price discovery.

We propose a new approach to measuring the contribution of trading venues to the price discovery process of internationally cross-listed stocks. We use a bivariate intensity approach as an alternative to the commonly applied vector error correction model, in order to take into account the irregularity of the data. Based on the autoregressive conditional intensity model developed by Russell (1999), contributions to price discovery are measured by modeling the interdependencies of the trading processes in both markets. In contrast to the Hasbrouck (1995) approach, the intensity-based information shares deliver unique results rather than upper and lower bounds. Furthermore, we show that the suggested approach provides a more accurate means to determine the dominant market in the price discovery process.

In our empirical application of this measure to the Canadian context we examine stocks which are listed on the TSX as well as on the NYSE. We find that trading on the TSX plays the most important role. This confirms previous results by Korczak and Phylaktis (2010), Eun and Sabherwal (2003), and Grammig et al. (2005), who also analyze Canadian stocks. We show that the leadership of the TSX is even more pronounced than indicated by previous studies. The average TSX information share amounts to 73%.

In order to explore potential determinants of a market's information share, we conduct cross-sectional regressions and like Grammig et al. (2005), Eun and Sabherwal (2003), and Korczak and Phylaktis (2010) find that liquidity contributes positively to a market's price determination.

Generally, our intensity-based unique information share presents an appealing alternative to the Hasbrouck (1995) information shares when analyzing price discovery using high frequency data, as it accounts for the irregular spacing of the data.

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Appendix A. Adjustment of intraday effects

In order to diurnally adjust the data, we follow Eubank and Speckman (1990) and regress the quote durations from the pooled process on polynomial and trigonometric time functions. The regression equation reads for some integers $d \geq 0$ and $\delta \geq 0$ as follows:

$$\tau_i = \beta_0 + \sum_{j=1}^d \beta_j^p t_i^j + \sum_{j=1}^{\delta} [\beta_j^c \cos(jt_i) + \beta_j^s \sin(jt_i)] + \epsilon_i, \quad (\text{A.1})$$

where $\tau_i = t_i - t_{i-1}$ gives the duration between consecutive price events. The number of polynomial and trigonometric terms are selected by a generalized cross-validation measure defined as,

$$\text{GCV} = \frac{n\text{RSS}}{(n - 2\delta - d - 1)^2}, \quad (\text{A.2})$$

where RSS denotes the residual sum of squares and n the number of observations. In order to avoid overfitting, we restrict d and δ to values smaller than three. To compute a typical time-of-day function, we select the specification of (A.1) that minimizes the GCV in (A.2).

Appendix B. Additional tables

See Tables 7 and 8.

Table 7
Stock-specific estimation results.

Ticker	$\hat{\omega}^1$	$\hat{\omega}^2$	$\hat{a}_1^1$	$\hat{a}_2^1$	$\hat{a}_1^2$	$\hat{a}_2^2$	$\hat{b}^1$	$\hat{b}^2$
ABX	−0.860 (0.027)	−0.678 (0.020)	0.082 (0.015)	0.050 (0.007)	0.029 (0.008)	0.078 (0.011)	0.932 (0.024)	0.909 (0.020)
ABY	−0.952 (0.049)	−0.560 (0.056)	0.126 (0.031)	0.055 (0.021)	0.048 (0.015)	0.084 (0.021)	0.853 (0.047)	0.880 (0.050)
AEM	−0.925 (0.030)	−0.702 (0.029)	0.110 (0.016)	0.058 (0.010)	0.056 (0.010)	0.100 (0.013)	0.834 (0.038)	0.852 (0.035)
AGU	−1.074 (0.053)	−0.909 (0.055)	0.098 (0.013)	0.050 (0.009)	0.036 (0.008)	0.076 (0.014)	0.916 (0.017)	0.901 (0.029)
AL	−0.940 (0.019)	−0.738 (0.022)	0.106 (0.007)	0.056 (0.006)	0.032 (0.004)	0.077 (0.008)	0.935 (0.008)	0.930 (0.014)
BCE	−1.079 (0.035)	−0.655 (0.039)	0.177 (0.021)	0.051 (0.009)	0.025 (0.008)	0.060 (0.010)	0.875 (0.025)	0.930 (0.018)
BCM	−1.268 (0.031)	−0.737 (0.045)	0.117 (0.014)	0.055 (0.010)	0.029 (0.005)	0.043 (0.006)	0.915 (0.016)	0.938 (0.016)
BEI	−0.936 (0.145)	0.011 (0.128)	0.243 (0.202)	−0.024 (0.010)	0.021 (0.038)	0.044 (0.016)	0.702 (0.359)	0.976 (0.016)
BMO	−1.378 (0.041)	−0.840 (0.049)	0.105 (0.014)	0.040 (0.007)	0.015 (0.004)	0.036 (0.006)	0.948 (0.011)	0.960 (0.009)
BNN	−1.604 (0.049)	−0.998 (0.052)	0.114 (0.010)	0.052 (0.007)	0.020 (0.004)	0.045 (0.007)	0.943 (0.008)	0.948 (0.010)
BNS	−1.435 (0.039)	−0.645 (0.052)	0.133 (0.017)	0.067 (0.016)	0.024 (0.005)	0.045 (0.007)	0.919 (0.017)	0.910 (0.029)

Table 7 (continued)

Ticker	$\hat{\omega}^1$	$\hat{\omega}^2$	$\hat{a}_1^1$	$\hat{a}_2^1$	$\hat{a}_1^2$	$\hat{a}_2^2$	$\hat{b}^1$	$\hat{b}^2$
BPO	−1.500 (0.056)	−0.752 (0.048)	0.113 (0.020)	0.034 (0.009)	0.025 (0.008)	0.102 (0.011)	0.914 (0.023)	0.889 (0.021)
BR	−2.304 (0.067)	−1.678 (0.033)	0.079 (0.004)	0.020 (0.004)	−0.023 (0.003)	0.095 (0.012)	0.987 (0.001)	0.960 (0.010)
BVF	−1.250 (0.043)	−1.143 (0.051)	0.081 (0.006)	0.050 (0.005)	0.035 (0.005)	0.069 (0.009)	0.976 (0.003)	0.970 (0.005)
CCJ	−1.747 (0.047)	−0.925 (0.057)	0.107 (0.010)	0.063 (0.006)	0.016 (0.004)	0.093 (0.009)	0.968 (0.004)	0.924 (0.013)
CGT	−0.658 (0.087)	−0.278 (0.080)	0.213 (0.042)	0.171 (0.047)	0.011 (0.017)	0.042 (0.021)	0.860 (0.048)	0.833 (0.091)
CJR	−2.135 (0.076)	−1.535 (0.113)	0.101 (0.033)	−0.034 (0.006)	−0.006 (0.014)	0.076 (0.009)	0.897 (0.049)	0.972 (0.004)
CLS	−1.031 (0.040)	−1.057 (0.048)	0.091 (0.009)	0.049 (0.007)	0.032 (0.005)	0.054 (0.009)	0.961 (0.006)	0.966 (0.008)
CNI	−1.200 (0.033)	−0.833 (0.035)	0.121 (0.009)	0.065 (0.007)	0.020 (0.004)	0.089 (0.010)	0.937 (0.007)	0.887 (0.023)
CNQ	−1.552 (0.042)	−0.779 (0.061)	0.137 (0.014)	0.062 (0.010)	0.023 (0.003)	0.071 (0.010)	0.941 (0.010)	0.926 (0.020)
COT	−1.462 (0.050)	−1.039 (0.047)	0.070 (0.008)	0.028 (0.006)	0.018 (0.005)	0.052 (0.008)	0.960 (0.006)	0.962 (0.007)
CP	−1.223 (0.040)	−0.774 (0.080)	0.117 (0.014)	0.057 (0.016)	0.025 (0.006)	0.049 (0.016)	0.918 (0.016)	0.925 (0.042)
CWG	−1.291 (0.075)	−1.266 (0.082)	0.054 (0.017)	0.010 (0.005)	0.025 (0.011)	0.026 (0.011)	0.937 (0.024)	0.973 (0.014)
DTC	−1.180 (0.054)	−0.715 (0.048)	0.095 (0.019)	0.055 (0.015)	0.034 (0.009)	0.084 (0.012)	0.918 (0.025)	0.877 (0.032)
ECA	−1.076 (0.033)	−0.871 (0.040)	0.105 (0.009)	0.051 (0.007)	0.032 (0.005)	0.040 (0.006)	0.951 (0.007)	0.962 (0.009)
ENB	−1.825 (0.051)	−0.855 (0.051)	0.137 (0.016)	0.059 (0.009)	0.016 (0.005)	0.057 (0.006)	0.940 (0.010)	0.932 (0.010)
ERF	−1.584 (0.050)	−1.282 (0.049)	0.087 (0.008)	0.043 (0.004)	0.031 (0.005)	0.045 (0.006)	0.969 (0.003)	0.977 (0.003)
FDG	−1.819 (0.058)	−1.630 (0.070)	0.093 (0.009)	0.049 (0.005)	0.022 (0.004)	0.056 (0.008)	0.967 (0.004)	0.970 (0.005)
FFH	−1.317 (0.045)	−1.533 (0.030)	0.145 (0.006)	0.015 (0.004)	−0.014 (0.004)	0.147 (0.008)	0.948 (0.004)	0.918 (0.007)
FHR	−1.155 (0.053)	−1.118 (0.053)	0.085 (0.012)	0.045 (0.007)	0.042 (0.007)	0.091 (0.010)	0.917 (0.016)	0.929 (0.012)
FS	−1.099 (0.040)	−1.088 (0.047)	0.090 (0.008)	0.071 (0.006)	0.027 (0.006)	0.159 (0.018)	0.931 (0.010)	0.845 (0.029)
GG	−0.909 (0.030)	−0.771 (0.035)	0.066 (0.009)	0.034 (0.005)	0.035 (0.006)	0.046 (0.008)	0.950 (0.010)	0.970 (0.007)
GIB	−0.954 (0.069)	−0.550 (0.085)	0.122 (0.024)	0.052 (0.022)	0.038 (0.015)	0.054 (0.021)	0.873 (0.029)	0.927 (0.044)
GIL	−1.825 (0.060)	−1.286 (0.066)	0.141 (0.014)	0.030 (0.006)	0.023 (0.007)	0.098 (0.007)	0.925 (0.009)	0.946 (0.006)
GLG	−1.111 (0.041)	−0.903 (0.049)	0.086 (0.009)	0.060 (0.008)	0.034 (0.006)	0.072 (0.012)	0.938 (0.011)	0.932 (0.018)
HBG	−1.107 (0.059)	−1.520 (0.092)	0.080 (0.007)	0.041 (0.010)	0.026 (0.009)	0.114 (0.033)	0.925 (0.009)	0.900 (0.041)
IDR	−1.170 (0.046)	−0.968 (0.050)	0.138 (0.014)	0.039 (0.009)	0.034 (0.009)	0.104 (0.011)	0.850 (0.023)	0.905 (0.018)
IPS	−1.389 (0.081)	−1.219 (0.074)	0.112 (0.029)	0.005 (0.007)	0.025 (0.010)	0.059 (0.009)	0.901 (0.040)	0.970 (0.006)
IQW	−1.521 (0.053)	−0.911 (0.060)	0.115 (0.015)	0.045 (0.009)	0.024 (0.009)	0.039 (0.011)	0.932 (0.013)	0.952 (0.015)
ITP	−1.386 (0.061)	−1.039 (0.066)	0.111 (0.012)	0.029 (0.008)	0.016 (0.008)	0.073 (0.009)	0.902 (0.014)	0.932 (0.010)
KFS	−1.486 (0.064)	−0.745 (0.058)	0.123 (0.016)	0.044 (0.010)	0.015 (0.007)	0.070 (0.009)	0.944 (0.009)	0.938 (0.010)
KGC	−0.786 (0.036)	−0.600 (0.033)	0.138 (0.022)	0.079 (0.015)	0.044 (0.013)	0.093 (0.024)	0.878 (0.033)	0.868 (0.051)
MDG	−1.063 (0.036)	−0.644 (0.028)	0.114 (0.014)	0.078 (0.013)	0.033 (0.008)	0.107 (0.017)	0.910 (0.018)	0.801 (0.057)
MDZ	−1.224 (0.055)	−0.766 (0.057)	0.099 (0.016)	0.076 (0.017)	0.013 (0.006)	0.041 (0.009)	0.914 (0.020)	0.877 (0.036)
MFC	−1.080 (0.033)	−0.839 (0.038)	0.134 (0.012)	0.086 (0.008)	0.037 (0.006)	0.113 (0.012)	0.902 (0.014)	0.882 (0.019)
MGA	−1.216 (0.031)	−0.908 (0.041)	0.116 (0.007)	0.051 (0.005)	0.010 (0.003)	0.097 (0.013)	0.935 (0.006)	0.890 (0.025)
MHM	−2.031 (0.070)	−1.270 (0.069)	0.134 (0.018)	0.023 (0.006)	0.017 (0.006)	0.081 (0.007)	0.935 (0.012)	0.952 (0.005)
MIM	−1.263 (0.073)	−1.434 (0.068)	0.114 (0.013)	0.050 (0.008)	0.033 (0.007)	0.114 (0.011)	0.903 (0.017)	0.926 (0.010)
MWI	−1.369 (0.082)	−1.386 (0.085)	0.089 (0.013)	0.055 (0.011)	0.052 (0.008)	0.120 (0.019)	0.961 (0.007)	0.931 (0.016)

(continued on next page)

Table 7 (continued)

Ticker	$\hat{\omega}^1$	$\hat{\omega}^2$	$\hat{a}_1^1$	$\hat{a}_2^1$	$\hat{a}_1^2$	$\hat{a}_2^2$	$\hat{b}^1$	$\hat{b}^2$
N	-0.899 (0.025)	-0.934 (0.033)	0.097 (0.008)	0.048 (0.005)	0.023 (0.004)	0.061 (0.007)	0.940 (0.009)	0.951 (0.009)
NCX	-1.137 (0.041)	-0.818 (0.043)	0.098 (0.014)	0.036 (0.007)	0.012 (0.006)	0.065 (0.008)	0.902 (0.020)	0.911 (0.017)
NRD	-1.213 (0.042)	-0.849 (0.063)	0.141 (0.015)	0.062 (0.012)	0.028 (0.006)	0.033 (0.010)	0.926 (0.013)	0.954 (0.018)
NT	-0.169 (0.071)	-0.486 (0.061)	0.154 (0.017)	0.053 (0.013)	0.132 (0.016)	0.178 (0.018)	0.970 (0.005)	0.963 (0.007)
NXY	-1.537 (0.041)	-1.032 (0.068)	0.113 (0.009)	0.046 (0.006)	0.022 (0.003)	0.063 (0.008)	0.945 (0.006)	0.947 (0.011)
OPY	-1.196 (0.097)	-1.552 (0.076)	0.126 (0.014)	0.023 (0.021)	-0.053 (0.010)	0.143 (0.034)	0.934 (0.009)	0.834 (0.061)
PCZ	-1.692 (0.047)	-1.152 (0.042)	0.129 (0.009)	0.060 (0.006)	0.023 (0.004)	0.036 (0.005)	0.968 (0.003)	0.975 (0.003)
PDG	-0.834 (0.027)	-0.710 (0.030)	0.072 (0.008)	0.038 (0.006)	0.054 (0.007)	0.086 (0.015)	0.947 (0.010)	0.943 (0.017)
PDS	-1.115 (0.036)	-1.041 (0.062)	0.145 (0.012)	0.048 (0.007)	0.024 (0.005)	0.070 (0.011)	0.896 (0.013)	0.931 (0.020)
PGH	-1.664 (0.062)	-1.270 (0.054)	0.094 (0.009)	0.054 (0.007)	0.035 (0.007)	0.045 (0.010)	0.981 (0.002)	0.984 (0.003)
PKZ	-1.435 (0.054)	-1.341 (0.065)	0.147 (0.010)	0.038 (0.005)	0.053 (0.006)	0.079 (0.009)	0.918 (0.009)	0.952 (0.008)
POT	-1.023 (0.037)	-1.156 (0.036)	0.131 (0.010)	0.022 (0.004)	0.009 (0.004)	0.132 (0.013)	0.897 (0.015)	0.887 (0.020)
PWI	-1.499 (0.056)	-1.366 (0.053)	0.084 (0.011)	0.056 (0.008)	0.034 (0.009)	0.040 (0.011)	0.979 (0.004)	0.984 (0.004)
RBA	-2.008 (0.083)	-1.459 (0.090)	0.103 (0.008)	0.027 (0.005)	0.051 (0.008)	0.124 (0.013)	0.957 (0.004)	0.955 (0.007)
RCN	-1.912 (0.068)	-1.467 (0.076)	0.188 (0.026)	0.031 (0.006)	0.025 (0.008)	0.100 (0.007)	0.899 (0.014)	0.939 (0.006)
RG	-1.368 (0.049)	-0.840 (0.053)	0.135 (0.014)	0.070 (0.011)	0.037 (0.008)	0.052 (0.007)	0.895 (0.017)	0.907 (0.018)
RY	-1.339 (0.033)	-0.706 (0.039)	0.118 (0.014)	0.054 (0.008)	0.013 (0.004)	0.033 (0.005)	0.944 (0.010)	0.951 (0.011)
RYG	-1.838 (0.075)	-1.401 (0.065)	0.131 (0.017)	0.066 (0.010)	0.043 (0.009)	0.052 (0.010)	0.952 (0.008)	0.964 (0.007)
SLF	-1.176 (0.033)	-0.751 (0.036)	0.127 (0.011)	0.089 (0.012)	0.033 (0.005)	0.066 (0.009)	0.914 (0.012)	0.892 (0.024)
SU	-1.074 (0.035)	-0.809 (0.041)	0.112 (0.010)	0.064 (0.008)	0.037 (0.006)	0.061 (0.010)	0.939 (0.010)	0.931 (0.017)
TAC	-1.308 (0.061)	-0.484 (0.055)	0.122 (0.026)	0.048 (0.014)	0.031 (0.009)	0.023 (0.006)	0.916 (0.026)	0.953 (0.015)
TD	-1.180 (0.033)	-0.698 (0.050)	0.120 (0.016)	0.066 (0.015)	0.015 (0.005)	0.046 (0.012)	0.931 (0.016)	0.925 (0.031)
TEU	-1.357 (0.049)	-0.785 (0.064)	0.130 (0.013)	0.083 (0.015)	0.018 (0.007)	0.063 (0.012)	0.923 (0.011)	0.894 (0.031)
TLM	-1.358 (0.031)	-0.732 (0.051)	0.128 (0.011)	0.064 (0.010)	0.025 (0.004)	0.070 (0.012)	0.917 (0.011)	0.915 (0.025)
TOC	-1.364 (0.066)	-0.824 (0.060)	0.138 (0.031)	0.071 (0.014)	0.036 (0.007)	0.049 (0.009)	0.900 (0.035)	0.921 (0.024)
TRA	-1.502 (0.118)	-1.251 (0.082)	0.051 (0.018)	0.006 (0.005)	0.020 (0.009)	0.079 (0.015)	0.973 (0.009)	0.964 (0.009)
TRP	-1.405 (0.051)	-0.611 (0.048)	0.204 (0.039)	0.078 (0.016)	0.054 (0.010)	0.050 (0.011)	0.894 (0.033)	0.938 (0.021)
TU	-1.515 (0.053)	-1.075 (0.050)	0.115 (0.014)	0.035 (0.005)	0.004 (0.004)	0.025 (0.003)	0.939 (0.011)	0.971 (0.003)
ZL	-0.838 (0.080)	-0.673 (0.063)	0.111 (0.023)	0.028 (0.009)	0.012 (0.009)	0.020 (0.005)	0.968 (0.010)	0.993 (0.004)

The table contains estimated parameters of the ACI model. Standard errors are reported in parentheses. For full company names see Table 1.

Appendix C. Hasbrouck information shares and adjustment coefficient ratio

According to the law of one price, prices in different trading venues that refer to the same underlying asset are cointegrated, meaning that they can only deviate from each other in the short-run. Assume that TSX and NYSE price changes, $\Delta p_t = p_t - p_{t-1}$, can be described by a bivariate vector error correction model (VECM),

$$\Delta p_t = \alpha \beta' p_{t-1} + \Gamma_1 \Delta p_{t-1} + \dots + \Gamma_{q-1} \Delta p_{t-q+1} + u_t, \quad (\text{A.3})$$

Table 8

Stock-specific intensity-based information shares.

Ticker	TSX							
	IIS^1	$SE(IIS)$	HIS_{low}^1	HIS_{up}^1	HIS_{mid}^1	$SE(HIS_{mid})$	Adj^1	$SE(Adj)$
ABX	63.7	6.7	19.6	87.9	53.7	1.5	59.2	2.3
ABY	63.0	13.0	58.9	95.2	77.0	1.5	81.2	1.7
AEM	53.4	7.1	11.7	77.3	44.5	1.7	44.3	2.6
AGU	64.1	8.4	26.4	81.6	54.0	2.3	54.2	2.7
AL	70.4	3.4	9.2	85.8	47.5	1.4	46.3	3.1
BCE	86.1	5.3	66.3	96.9	81.6	2.2	84.8	2.7
BCM	83.8	3.0	50.9	99.5	75.2	1.7	91.9	3.4

Table 8 (continued)

Ticker	TSX							
	IIS^1	$SE(IIS)$	HIS_{low}^1	HIS_{up}^1	HIS_{mid}^1	$SE(HIS_{mid}^1)$	Adj^1	$SE(Adj)$
BMO	88.5	3.0	65.4	99.7	82.5	1.3	94.6	2.1
BNN	86.3	3.1	56.8	97.5	77.1	1.7	84.0	2.4
BNS	89.1	2.8	57.1	99.7	78.4	1.1	94.7	1.6
BPO	60.6	9.7	79.8	100.0	89.9	0.5	100.0	0.3
BVF	63.0	5.2	14.6	84.9	49.8	1.6	51.1	2.6
CCJ	82.0	3.3	41.5	96.7	69.1	1.8	79.1	2.8
CGT	98.7	2.4	65.5	98.8	82.2	2.0	89.6	3.0
CLS	72.3	4.5	32.7	90.4	61.5	1.8	68.1	2.3
CNI	81.4	3.5	15.6	92.1	53.9	27.7	52.7	17.0
CNQ	83.8	2.8	38.4	98.0	68.2	1.4	82.8	2.4
COT	67.0	7.6	16.4	66.1	41.3	1.5	41.4	1.4
CP	84.7	4.4	56.6	98.1	77.4	1.4	86.4	2.0
CWG	46.4	15.1	17.8	37.7	27.7	6.4	10.0	0.9
DTC	64.2	9.3	67.3	97.9	82.6	2.1	86.9	2.9
ECA	80.8	3.5	34.6	94.0	64.3	2.1	70.9	3.2
ENB	89.8	2.8	72.2	98.2	85.2	1.7	91.9	1.2
ERF	72.2	4.8	16.0	73.5	44.7	2.7	49.8	2.6
FDG	79.1	4.1	14.9	89.0	52.0	1.5	55.4	3.0
FHR	50.0	7.5	14.9	73.8	44.3	2.1	44.2	2.6
FS	59.8	5.2	1.3	57.3	29.3	1.3	13.6	1.7
GG	57.9	7.4	14.1	81.4	47.7	1.6	50.0	2.4
GIB	75.3	12.7	73.3	99.1	86.2	1.4	91.8	2.0
GIL	65.2	8.2	46.4	91.8	69.1	3.2	69.7	3.6
GLG	67.8	5.4	10.0	75.3	42.6	1.5	39.4	2.4
HBG	52.7	11.3	3.0	33.2	18.1	4.4	12.1	4.6
IDR	60.4	8.5	17.9	67.2	42.5	1.8	41.4	1.8
IPS	27.8	25.2	54.3	98.3	76.3	2.1	85.1	3.2
IQW	85.0	5.5	41.2	81.0	61.1	4.1	60.8	2.2
ITP	73.4	12.0	36.6	92.1	64.3	2.9	71.4	3.7
KFS	83.9	8.2	75.3	98.5	86.9	1.8	90.6	2.3
KGC	73.0	9.4	70.7	100.0	85.3	0.6	100.0	0.5
MDG	71.3	6.6	11.0	77.1	44.0	1.2	43.3	1.8
MDZ	93.4	3.6	55.5	99.4	77.4	1.7	90.9	3.1
MFC	73.6	4.2	31.5	84.5	58.0	4.5	60.8	5.0
MGA	86.5	4.5	15.9	79.1	47.5	2.3	46.1	3.2
MHM	68.9	8.0	50.5	95.9	73.2	2.3	79.4	3.1
MIM	59.7	7.2	2.9	56.8	29.9	1.8	19.0	2.3
MWI	44.0	8.3	1.5	38.1	19.8	2.6	9.2	1.7
N	76.4	3.7	25.3	96.3	60.8	0.8	73.1	1.7
NCX	82.3	8.5	40.9	91.0	65.9	1.9	69.0	2.2
NRD	90.5	2.8	69.3	99.2	84.3	1.4	91.3	2.0
NT	25.7	7.2	14.3	88.7	51.5	1.6	50.0	3.2
NDY	79.3	3.6	79.1	100.0	89.6	0.8	99.9	0.7
PCZ	90.1	2.0	53.1	99.9	76.5	1.1	96.8	1.9
PDG	36.9	6.5	22.8	90.1	56.4	1.7	62.9	2.7
PDS	80.7	4.4	33.9	91.1	62.5	1.9	67.6	2.5
PGH	76.5	6.3	37.8	81.2	59.5	3.0	61.9	2.2
PKZ	56.7	4.4	19.2	87.3	53.3	1.8	56.7	2.8
POT	71.2	11.7	12.6	79.3	46.0	1.3	44.3	2.0
PWI	78.0	7.0	26.0	87.1	56.5	2.3	62.8	2.7
RBA	30.7	5.7	1.7	21.7	11.7	1.8	12.0	2.2
RCN	70.0	7.5	28.5	82.4	55.5	3.1	56.7	3.4
RG	83.0	4.2	60.4	100.0	80.2	1.5	98.8	2.0
RY	93.7	2.0	65.7	99.6	82.6	1.7	93.5	2.8
RYG	79.4	4.8	53.5	99.3	76.4	1.8	90.4	3.0
SLF	83.8	3.1	55.8	99.5	77.6	2.0	91.9	3.7
SU	76.4	4.2	50.7	97.1	73.9	1.7	81.1	2.4
TAC	88.8	4.4	66.3	98.2	82.3	1.8	88.0	2.4
TD	91.8	2.9	68.9	99.9	84.4	1.5	96.5	2.3
TEU	90.4	3.7	46.7	96.2	71.5	2.1	79.5	2.9
TLM	82.0	3.1	33.0	95.7	64.4	2.4	75.3	4.0
TOC	84.8	3.6	59.8	99.6	79.7	1.5	92.8	2.7
TRA	17.1	14.4	0.0	1.1	0.6	0.5	0.2	0.4
TRP	85.6	3.5	74.4	99.0	86.7	1.8	92.4	1.8
TU	97.6	2.5	82.4	99.9	91.2	1.0	98.2	1.2
ZL	92.7	6.3	60.5	95.4	78.0	1.9	80.3	2.2

The table presents the stock-specific TSX intensity-based information share (IIS^1), the Hasbrouck information share midpoint together with lower and upper bounds (HIS_{mid}^1 , HIS_{low}^1 and HIS_{up}^1), and the adjustment coefficient ratio (Adj^1) in percent. The corresponding standard errors are reported in the columns labeled $SE(IIS)$, $SE(HIS_{mid}^1)$ and $SE(Adj)$. NYSE information shares can be calculated using $IIS^2 = 100 - IIS^1$, $HIS_{low}^2 = 100 - HIS_{up}^1$, $HIS_{up}^2 = 100 - HIS_{low}^1$, $HIS_{mid}^2 = 100 - HIS_{mid}^1$ and $Adj^2 = 100 - Adj^1$. The information shares are computed for 73 stocks that have positive α^s estimates. For full company names see Table 1.

where $p_t = (p_t^1, p_t^2)'$ and Γ_1 to Γ_{q-1} are 2×2 parameter matrices. $u_t = (u_t^1, u_t^2)'$ is a white noise vector with zero means and covariance matrix Σ_u . β denotes the 2×1 cointegration vector. The vector $\alpha = (\alpha^1, \alpha^2)'$ contains the coefficients associated with the speed of adjustment of each price series to deviations from the equilibrium. The ratio of these coefficients is frequently used in price discovery studies (see Booth et al., 1999; Harris et al., 2002; Eun and Sabherwal, 2003). The adjustment coefficient ratios given by

$$Adj^1 = \frac{|\alpha^2|}{\alpha^1 + |\alpha^2|} \quad \text{and} \quad Adj^2 = \frac{\alpha^1}{\alpha^1 + |\alpha^2|}, \quad (A.4)$$

measure the relative price adjustment in a market to deviations from the equilibrium. The more a market adjusts, the less it contributes to the price discovery process. By Adj^1 we denote the adjustment of NYSE prices reflecting TSX's contribution to price discovery. This holds vice versa for Adj^2 . However, focusing merely on the adjustment of prices and neglecting the innovations' variances and contemporaneous dependence structure has been widely criticized (see Hasbrouck, 1995; Hasbrouck, 2002; De Jong, 2002; Lehmann, 2002; Baillie et al., 2002).

The cointegration vector β in (A.3) implies that there exists one common stochastic trend, which, according to Hasbrouck (1995), can be considered as the stock's underlying efficient price. Hasbrouck's information shares are then derived as the contribution of an innovation in one market's price series to the efficient price innovations' variance. Since the VECM innovations, u_t , tend to be contemporaneously correlated, the shares cannot be uniquely identified. To solve this dilemma, Hasbrouck applies a Cholesky decomposition to the covariance matrix of innovations, $\Sigma_u = CC'$.

With the home market ordered first, Hasbrouck information shares of the TSX (HIS^1) and the NYSE (HIS^2) can be computed as,

$$HIS^1 = \frac{([\xi' C]_{[1]})^2}{\xi' C C' \xi} \quad \text{and} \quad HIS^2 = \frac{([\xi' C]_{[2]})^2}{\xi' C C' \xi}, \quad (A.5)$$

where $[\xi' C]_{[j]}$ denotes the j^{th} element of the vector $\xi' C$. ξ gives the common row vector in the matrix Ξ which contains the long-run impacts of time t idiosyncratic innovations on the efficient price. Ξ is given by

$$\Xi = \beta_{\perp} \left[\alpha'_{\perp} \left(I_n - \sum_{i=1}^{q-1} \Gamma_i \right) \beta_{\perp} \right]^{-1} \alpha'_{\perp}. \quad (A.6)$$

The Cholesky decomposition implies that the contribution of the market ordered first is maximized and that of the market ordered second is minimized. Since there is no theoretical justification for such a hierarchy, the common solution is to permute the ordering of the markets. This yields upper and lower bounds of information shares denoted by HIS_{up}^s and HIS_{low}^s , respectively. The main drawback of Hasbrouck's methodology is that these bounds can diverge considerably, as the contemporaneous correlation between the composite innovations u_t^1 and u_t^2 tends to increase with decreasing sampling frequency.

In our application to Canadian stocks we choose a one-minute sampling frequency using quoted prices and convert the US stock prices to Canadian Dollars. After testing for cointegration using the maximum eigenvalue and trace statistic, we are able to confirm the existence of one cointegration relation. For estimating the VECM we exclude overnight returns. The number of lags, q , on the right hand side in the VECM in (A.3) is determined by the Schwarz information criterion (SIC) (see Schwarz, 1978). The first q returns on each trading day are initialized by using the stock's average return over the whole sample. Standard errors for the adjustment coefficient ratios in (A.4) and Hasbrouck's information shares in (A.5) are derived using a nonparametric bootstrap as proposed by Grammig et al. (2008).

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